

745 **Appendix A. Supplementary Dataset Card**

746 To address the transparency of the dataset construc-
747 tion process and provide a comprehensive overview of la-
748 bel quality and geographic representativeness, we present
749 a supplementary dataset card for the EVLAB-HCR and
750 GID-HCR benchmarks in Table [A.8](#). This card details the
751 annotation protocols, quality control metrics, handling of
752 class imbalance, and regional limitations.

Table A.8: Dataset Card for EVLAB-HCR and GID-HCR

Category	Detailed Description
Data Source & Foundation	EVLAB-HCR: Built upon high-resolution imagery (0.8–2.0 m) and high-precision vector labels from China’s 2020 national geoinformation monitoring program. GID-HCR: Repurposed from the established large-scale public Gaofen Image Dataset (GID-5 and GID-15). Both datasets leverage authoritative data sources rather than starting from scratch via crowdsourcing.
Annotation Protocols & Inter-Annotator Agreement	EVLAB-HCR: Labels were not generated through ad hoc manual drawing. They were constructed using a systematic three-stage workflow: (1) knowledge-guided sampling using FROM-GLC10 and Landscape Shape Index; (2) acquisition of high-precision imagery and corresponding vector labels; and (3) hierarchical benchmark generation based on the industry standard CH/T 9029-2019. The high reliability of Level-3 labels is intrinsically guaranteed by the rigorous upstream national-quality production process (field investigation, standardized office processing, and strict quality inspection), thus a dedicated inter-annotator agreement statistic is not reported. GID-HCR: The annotation protocols strictly follow those of the original public GID dataset, which were meticulously annotated and cross-checked by professional remote sensing experts. The hierarchical refinement relies on the deterministic reorganization of the already-verified GID-15 fine-grained categories, inherently inheriting its established inter-annotator reliability without requiring supplementary manual relabeling.
Automated Quality Control	Both datasets: Automated topological and semantic consistency scripts were executed across all samples to guarantee that all fine-grained sub-categories strictly nested within their respective parent categories, ensuring zero logical violations across the hierarchy.
Class Imbalance Handling	Both datasets: The datasets exhibit a natural long-tailed class distribution (e.g., the limited number of <i>shrub land</i> test samples in GID-HCR). This imbalance genuinely reflects the real-world physical distribution of land cover in the sampled regions. To prevent the model from biasing toward majority classes, our CG-HCAN framework explicitly incorporates a composite loss function combining Tversky, weighted Cross-Entropy, and Focal losses during training.
Geographic Representativeness & Bias	Both datasets: Constructed using imagery geographically concentrated in China. While they cover diverse topographies, this China-centric distribution introduces a potential geographic bias when scaling to fundamentally different environments (e.g., tropical landscapes or arid zones). We explicitly acknowledge this limitation; achieving robust cross-domain adaptation across globally distributed datasets remains a primary focus of our ongoing and future research.